

Sayantan Roy

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EDUCATION

Purdue University , West Lafayette, IN, USA	2021–present
M.S. in Economics	2024
Ph.D. Candidate in Economics	May 2027 (expected)
Ph.D. committee: Mario J. Crucini (chair), Seungki Hong, Cathy M. Zhang	
Madras School of Economics , Chennai, Tamil Nadu, India	2017–2019
M.A. in Economics	
R.K.M.R. College, Narendrapur , Kolkata, West Bengal, India	2014–2017
B.S. in Economics	

RESEARCH FIELDS

Primary: Macroeconomics **Secondary:** International Trade

JOB MARKET PAPER

Where Does Fiscal Stimulus Create Jobs? Evidence from U.S. Counties

Abstract. A national stimulus program jointly sets the size of total funding and its allocation across states and counties. I study whether the latter matters for the employment the program creates. Using county-level variation in American Recovery and Reinvestment Act (2009–12) outlays, I find that the local employment responses to government spending are largest in mid-sized counties (about 100,000 residents) and substantially smaller in both small and large counties (about 10,000 and 1 million residents). Evidence on local production structure, commuting and purchasing patterns, and hiring points to two opposing forces. While small counties leak most of the demand they receive to other places, large counties turn the demand they retain into fewer jobs, consistent with greater worker bargaining power and higher recruiting costs. Mid-sized counties balance the two. I interpret and rationalize these patterns using a multi-region New Keynesian model in which locations are linked through trade, and labor markets feature search and matching frictions. In counterfactual reallocations with the aggregate budget unchanged, I find that moving 35 percent of large-county spending to mid-sized counties would have created about 145,800 additional job-years over 2009–12 (3.3 percent) and lowered the cost per job-year from \$48,109 to \$46,551. The spatial allocation of stimulus is thus an important dimension of stabilization policy distinct from the size of the program.

WORK IN PROGRESS

Tariffs Along the Supply Chain: Production Networks and Aggregate Adjustment in the Interwar United States

with Mario J. Crucini

Common Shocks, Local Exposure: Sectoral Wage Dynamics Across U.S. Counties

Gender Gaps in Employment Seasonality: The Role of Unemployment Insurance

RESEARCH EXPERIENCE

Research Assistant , Mario J. Crucini, Purdue University	2023–present
Research Assistant , Seungki Hong, Purdue University	2022

TEACHING EXPERIENCE

Instructor of Record , Purdue University Introductory Macroeconomics (ECON 252)	Summer 2023, 2025
Teaching Assistant , Purdue University Macroeconomic Theory III (ECON 612, Ph.D.); Money and Banking (ECON 380); Introductory Econometrics (ECON 360); Intermediate Macroeconomics (ECON 352)	2021–present

INDUSTRY EXPERIENCE

Data Scientist , American Express Credit and Fraud Risk Division	2019–2021
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AWARDS AND FELLOWSHIPS

Krannert Doctoral Students' Research Symposium Winner, Purdue University	2024
Novshek-Watts Scholarship, Purdue University	2021
Graduate Excellence Award, Madras School of Economics	2019
Academic Fellowship, Madras School of Economics	2017–2019
Academic Excellence Award, R.K.M.R. College, Narendrapur	2017

CONFERENCE PRESENTATIONS

Midwest Macroeconomics Meetings, Marquette University	2026
20th Annual Conference on Economic Growth and Development, ISI Delhi	2025

COMPUTATIONAL SKILLS

Python, JAX, Julia, MATLAB, Stata, SQL, PySpark, Git

REFERENCES

Mario J. Crucini Professor Department of Economics Purdue University mcrucini@purdue.edu	Seungki Hong Assistant Professor Department of Economics Purdue University hong397@purdue.edu	Cathy M. Zhang Professor Department of Economics Purdue University cmzhang@purdue.edu
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